



Bus Ticket Reservation System

A Full Stack Application

Java

Spring Boot

MySQL

ReactJS

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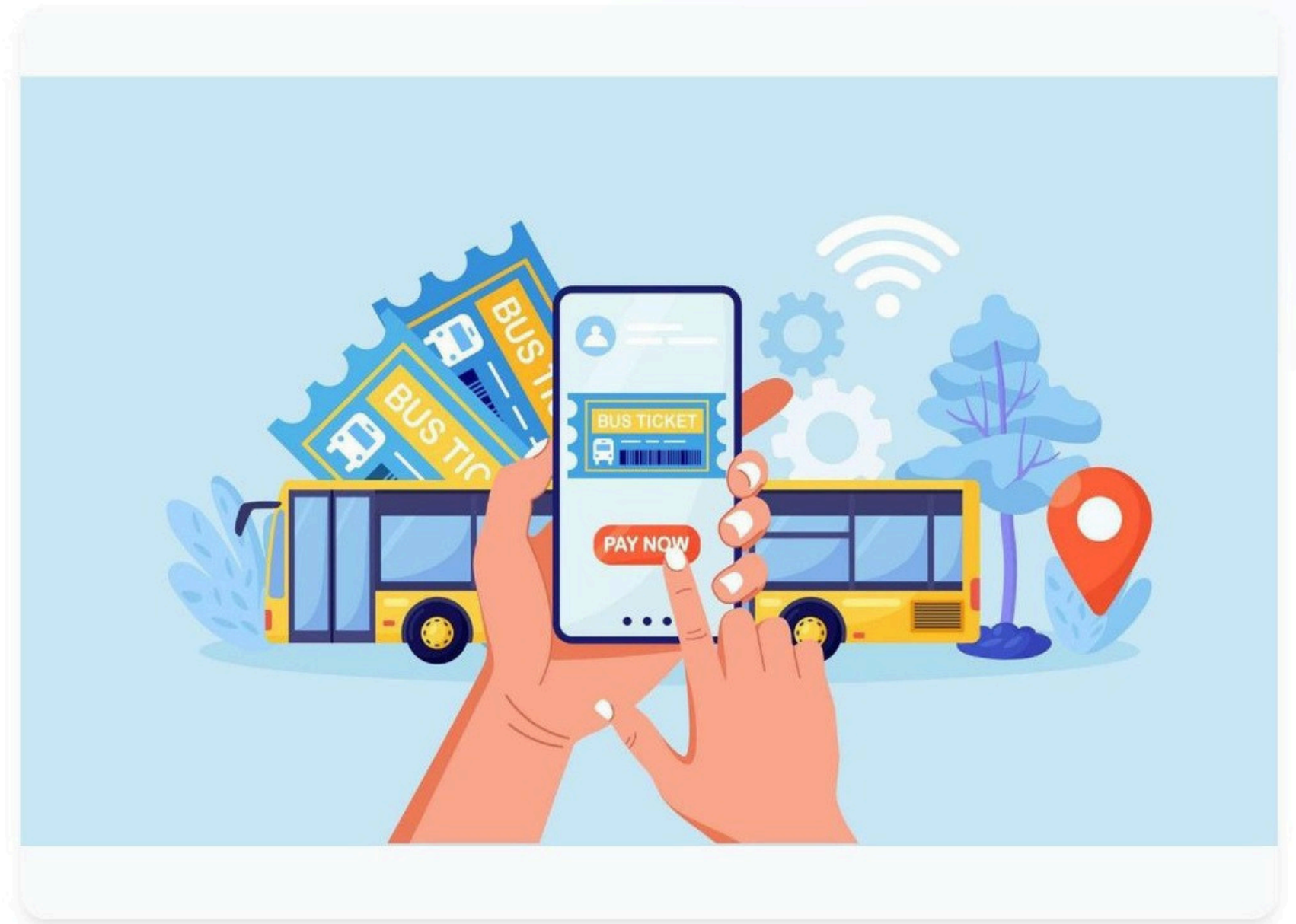
Professor & HOD

Introduction

Digitalizing the Journey

The transition from manual booking counters to digital systems is essential for modern transportation. This project aims to bridge that gap.

- ✓ Shift from slow, manual processes to instant online actions.
- ✓ Requirement for fast search, booking, and immediate cancellation.
- ✓ Provides a unified, efficient platform for travelers.



Problem & Solution

! Existing Problems

- Manual booking consumes excessive time for both staff and passengers.
- Complex management of fluctuating schedules and dynamic fare pricing.
- Lack of organized booking history leads to data loss.
- Admin workload is unmanageable without automation.

💡 Proposed Solution

- A robust, web-based reservation system backed by a relational database.
- Streamlined interface with distinct roles for Admins and Users.
- Automated fare calculation and real-time seat availability.
- Centralized data management for easy reporting.

Project Objectives

-  **Online Access:** Provide a 24/7 platform for booking tickets from anywhere.
-  **Search Functionality:** Enable users to find buses by specific travel dates and routes.
-  **Cancellation:** Allow hassle-free booking cancellation with status updates.
-  **Tracking:** Maintain detailed records using unique booking IDs for every transaction.
-  **Admin Control:** Empower admins to manage fleets, schedules, and pricing dynamically.

Roles & Permissions



User Role

- Register & Login securely
- Search buses by source, destination, date
- Book tickets & View booking history
- Cancel tickets via Booking ID



Admin Role

- Full access to user actions
- Add, Remove, or Modify Bus details
- Update journey dates and Fares
- View all system-wide bookings

Functional Modules



Authentication Module:

Secure login/registration for Users and Admins.



Bus Search Module:

Filtering logic based on travel dates and routes.



Booking Module:

Ticket generation, seat allocation, and ID creation.



Cancellation Module:

Logic to revert bookings and update seat status.



History Module:

User-specific dashboard for past and upcoming trips.



Bus Management (Admin):

CRUD operations for the bus fleet and schedules.



Booking Management (Admin):

Oversight of all reservations in the system.

Technology Stack



Frontend

ReactJS

CSS3 / Styled Components



Backend

Java

Spring Boot (REST APIs)



Database

MySQL

Relational Data Model



Tools

VS Code, IntelliJ/Eclipse

Postman (API Testing)

System Architecture

3-Tier Architecture Flow



Presentation Layer

ReactJS handles user interaction and displays data to the client.



Business Logic

Spring Boot processes API requests and manages core application rules.



Data Layer

MySQL securely stores persistent data like users, buses, and bookings.

Data Flow: User ➡ Frontend (React) ➡ Backend (Spring Boot) ➡ Database (MySQL)

Database Design

Users Table

- user_id (PK)
- name
- email
- password
- role (ENUM)

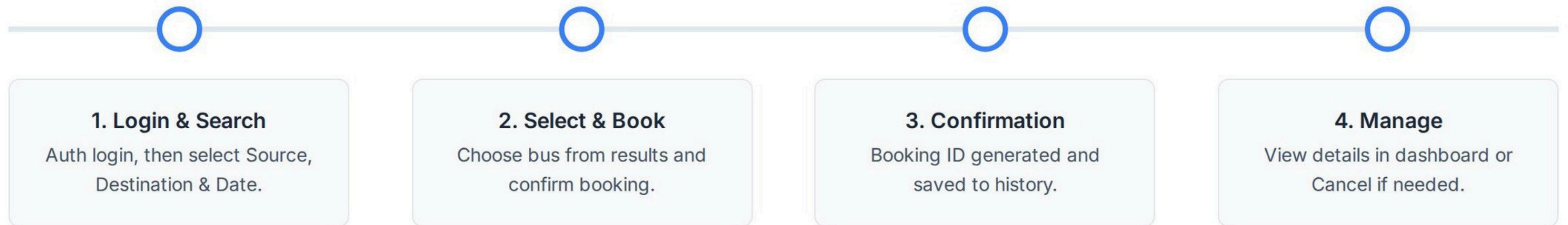
Buses Table

- bus_id (PK)
- bus_name
- source / dest
- journey_date
- fare

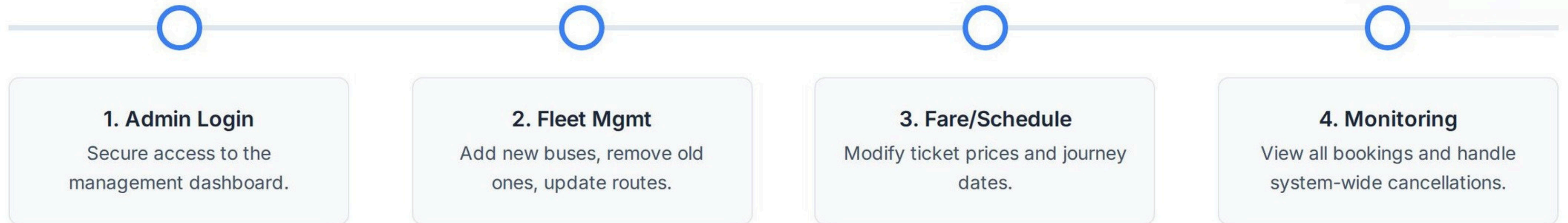
Bookings Table

- booking_id (PK)
- user_id (FK)
- bus_id (FK)
- status
- booking_date

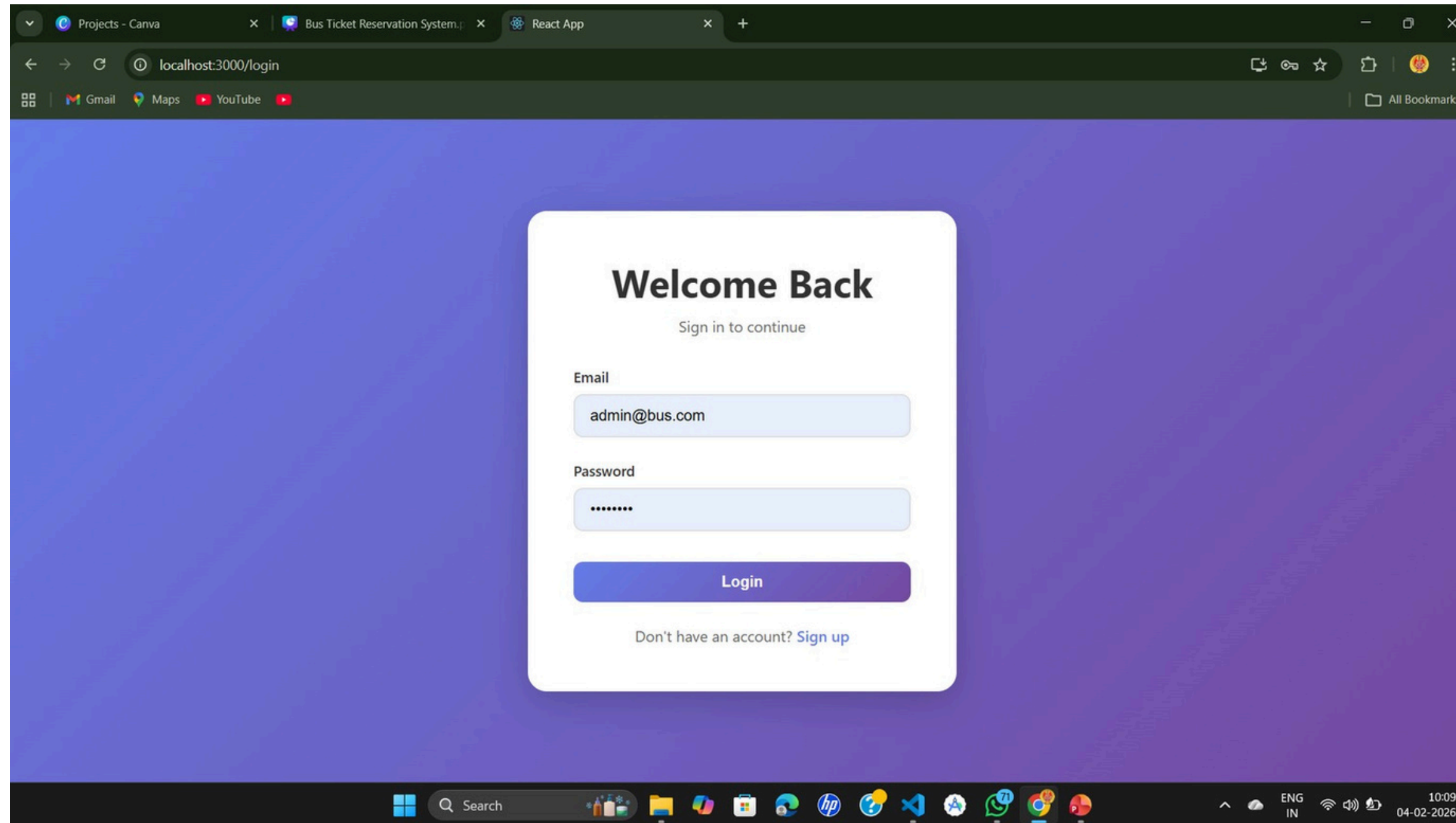
User Workflow



Admin Workflow



Output Login Screen



Admin's Interface

Blue Bus

HomeBusesMy BookingsAdmin

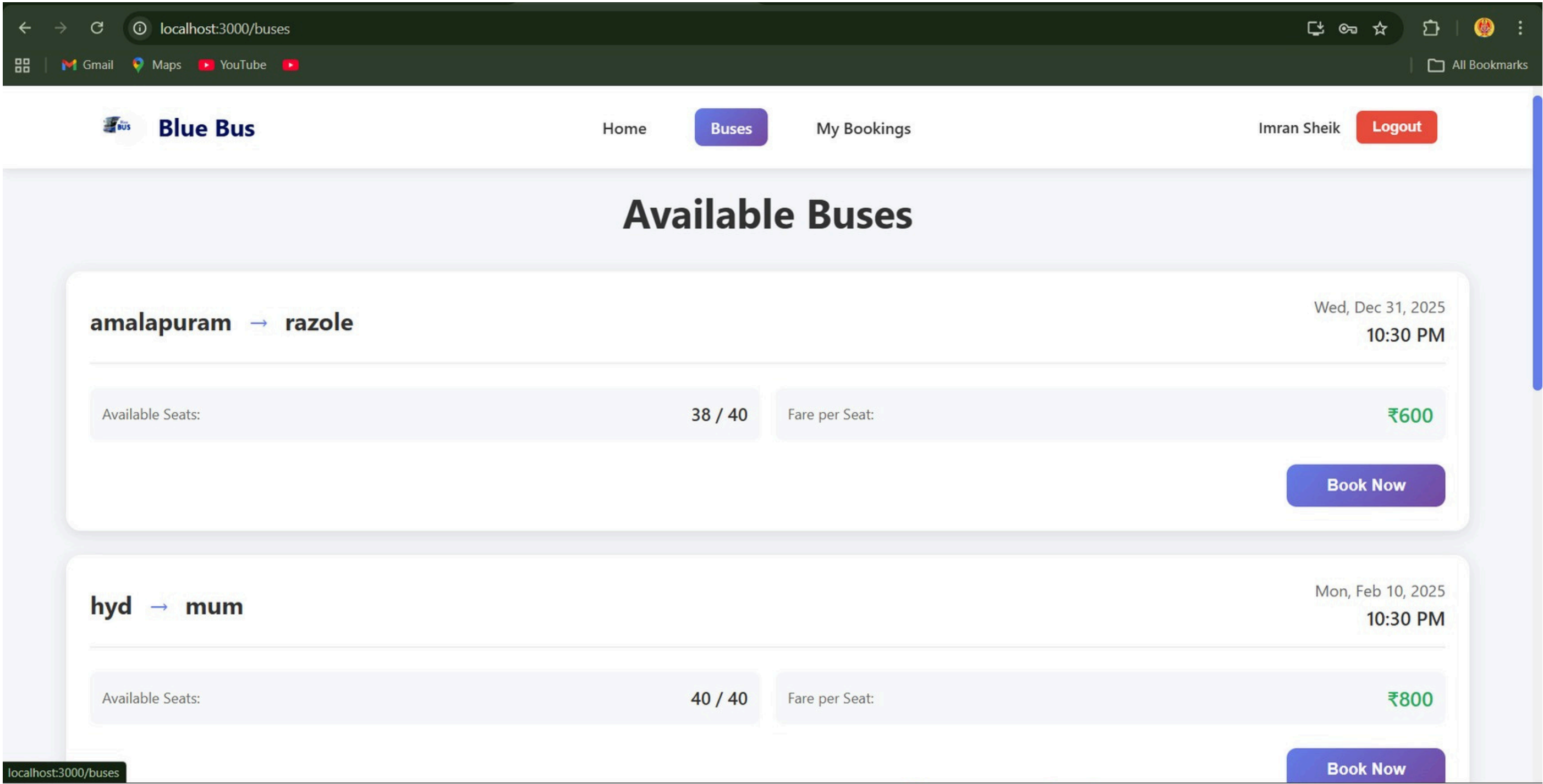
Admin UserLogout

Bus Management

+ Add New Bus

Route	Date	Time	Seats	Available	Fare	Actions
amalapuram → razole	Dec 31, 2025	10:30 PM	40	38	₹600	EditDelete
hyd → mum	Feb 10, 2025	10:30 PM	40	40	₹800	EditDelete
razole → rajamundry	Dec 28, 2025	11:30 AM	40	40	₹400	EditDelete
Banglore → Srikakulam	Dec 30, 2025	12:20 PM	30	30	₹500	EditDelete
amalapuram → razole	Dec 30, 2025	10:30 AM	48	48	₹120	EditDelete

User's Bus Availability Interface



My Bookings Section

Blue Bus

[Home](#)[Buses](#)[My Bookings](#)

Imran Sheik [Logout](#)

Booking ID: #11

CANCELLED

Journey Details		Passenger Information		Booking Information	
Route:	amalapuram → razole	Name:	Imran Sheik	Seats:	5
Journey Date:	Wed, Dec 31, 2025	Email:	imransheik@gmail.com	Total Fare:	₹3000
Journey Time:	10:30 PM	Phone:	9182969582	Booking Date:	Dec 28, 2025, 11:47 AM
				Cancelled On:	Dec 28, 2025, 11:48 AM

Booking ID: #12

CONFIRMED

Journey Details		Passenger Information		Booking Information	
Route:	amalapuram → razole	Name:	Imran Sheik	Seats:	1
Journey Date:	Wed, Dec 31, 2025	Email:	imransheik@gmail.com	Total Fare:	₹600
Journey Time:	10:30 PM	Phone:	9182969582	Booking Date:	Jan 5, 2026, 11:35 AM

Cancel Booking

Booking ID: #13

CONFIRMED

Conclusion & Future Scope



🚩 Conclusion

The system successfully simplifies booking with a user-friendly React UI and robust Spring Boot backend, ensuring efficient management through unique Booking IDs.

🔧 Future Enhancements

Seat Selection: Visual seat map for users.

Payment Gateway: Real-time online payments.

Notifications: SMS/Email confirmation.

PDF Ticket: Downloadable e-tickets.